CO-2 _ AT-01

Analytical problem solving

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Course :- Computer vision.

1. Grey level transformation

Grey level transformation in image

a) purpose of Grey level enhancement

→ used to change the intensity values of pixels in a grayscale image.

* Enhance image contrast
* Highlights important features.
* Improve Image visibility.

b) apply negative transformation.

negative transformation :- $\boxed{s = L - 1 - r}$

r — original pixel value.

s — Transformed pixel value

L - 256

$\therefore s = 256 - 1 - r = 255$

Examples :- suppose original pixels values.

original pixel values (r)	Negative pixel value ($s = 255 - r$)
0	255
50	205
100	155
150	105
200	55

Histogram Equilization

a) Image enhancement technique used to increase the contrast of an image spreading the intensity values more evenly over the original range.

Purpose :— — Improve image contrast
— Enhance visibility of hidden details
— Spreads pixels uniformly.

Total no. of pixels (N) :–
$N = 4+6+10+6 = 26$.

b) Cumulative Distribution function (CDF)

Intensity	frequency	$P(r_k) = \frac{n_k}{26}$	CDF
0	4	$4/26 = 0.156$	0.154
1	6	$6/26 = 0.231$	0.154 + 0.231 =
2	10	$10/26 = 0.385$	0.385 + 0.38
3	6	$6/26 = 0.231$	0.770 + 0.231 =

c) Histogram equilization formula.

$s_k = (L-1) \times CDF$

for a 4 bit image, $L = 16$ so $\boxed{L-1 = 15}$

Intensity	CDF	15×CDF	Equilised Intensity
0	0.154	2.31	2
1	0.385	5.78	6
2	0.770	11.55	12
3	1.000	15.00	15

Gaussian smoothing

a) More importance to the center pixel and less importance to neighbouring pixels, resulting in smoother images while preserving edges better than an average filtering

advantages :- reduces random noise effectively
Preserves image edges better

b) Apply Gaussian kernel.

3×3 Gaussian kernel : $\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

New center pixel = $\dfrac{\Sigma(\text{Pixel} \times \text{kernel weight})}{16}$

Image region : $\begin{bmatrix} 120 & 125 & 130 \\ 118 & 122 & 128 \\ 115 & 120 & 126 \end{bmatrix}$

Step 1 :- $(120\times1) + (125\times2) + (130\times1) + (118\times2) + (122\times4)$
$+ (120\times2) + (126\times1)$
$+ (128\times2) + (115\times1) +$

Step 2 :- $120 + 250 + 130 + 236 + 488 + 256 + 115 + 240$
$+ 126 = 1961$

Divide by 16 $= \frac{1961}{16} = 122.56$

$\boxed{\text{answer} = 122.56 = 123}$

(4) Edge detection using sobel operator

a) Edge detection identifies the boundaries in image by detecting sudden changes in pixel intensity.

b) apply the Horizontal sobel mask

$1 \longrightarrow \quad G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$

$2 \longrightarrow \quad \text{Image region} = \begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$.

$3 \longrightarrow \quad (-1 \times 10) + (0 \times 20) \quad \ldots + (0 \times 40) + (50 \times 1)$

$4 \longrightarrow \quad -10 + 0 + 30 - 40 + 0 + 80 - 30 + 0 + 50$

$\Rightarrow 80.$

Gradiant at the center = 80.

Contrast streching

a) Need for contrast streching.

contrast streching is an image enhancement technique used to increase the contrast of an image by expanding the range of intensity values.

b) Linear constrast streching.

Formula :- $s = \frac{(r - r_{min})}{(r_{max} - r_{min})} \times 255$

$r_{min} = 50$

$r_{max} = 150$.

50,

$s = \frac{(r-50)}{100} \times 255$

Examples :- Intensity values 50, 75, 100, 125, 150.

original intensity	calculation	Transformd.
50	$(50-50)/100 \times 255)$	0
75	$(70-50)/100 \times 255 = 63.75$	64
100	127.5	128
125	191.25	191
150	255	255